

CS 236: Scientific Computing

Problem Set 1

Rayo, Joshua Frankie
jbrayo@up.edu.ph

Ricio, Dustin Edric
dlricio@up.edu.ph

March 17, 2016

Contents

1	Interpolation	2
1.1	Problem 7.5	2
1.2	Problem 7.12	6
2	Numerical Integration and Differentiation	14
2.1	Problem 8.1	14
2.2	Problem 8.2	15
2.3	Problem 8.4	16
2.4	Problem 8.6	18
3	Initial Value Problems for ODEs	20
3.1	Problem 9.2	20
3.2	Problem 9.4	21
3.3	Problem 9.8	23
3.4	Problem 9.9	25

1 Interpolation

1.1 Problem 7.5

(a) Determine the polynomial interpolant using monomial basis to the data below.

t	1	2	3	4
y	11	29	65	125

Solution. There is a unique polynomial

$$p_3(t) = x_0 + x_1t + x_2t^2 + x_3t^3$$

that interpolates the given four points. Using the monomial basis, we need to solve x_i s such that

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 11 \\ x_1 + 2x_2 + 4x_3 + 8x_4 &= 29 \\ x_1 + 3x_2 + 9x_3 + 27x_4 &= 65 \\ x_1 + 4x_2 + 16x_3 + 64x_4 &= 125 \end{aligned}$$

This system of equations will lead to the matrix form

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 11 \\ 29 \\ 65 \\ 125 \end{bmatrix}$$

$$\mathbf{Ax} = \mathbf{y}$$

Where $\mathbf{A}$ is a Vandermonde matrix. This guarantees that the system will have a solution if different values of t are used. One can obtain the solution using Gauss-Jordan reduction method. Perform elementary row operations on the augmented matrix $[\mathbf{A}|\mathbf{y}]$ until $\mathbf{A}$ becomes $\mathbf{I}$. Solving the system, we have

$$\begin{aligned} &\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 1 & 2 & 4 & 8 & 29 \\ 1 & 3 & 9 & 27 & 65 \\ 1 & 4 & 16 & 64 & 125 \end{array} \right] \xrightarrow{\substack{R_2-R_1 \\ R_3-R_1 \\ R_4-R_1}} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 0 & 1 & 3 & 7 & 18 \\ 0 & 2 & 8 & 26 & 54 \\ 0 & 3 & 15 & 63 & 114 \end{array} \right] \xrightarrow{\substack{R_3-2R_2 \\ R_4-3R_2}} \\ &\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 0 & 1 & 3 & 7 & 18 \\ 0 & 0 & 2 & 12 & 18 \\ 0 & 0 & 6 & 42 & 60 \end{array} \right] \xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 0 & 1 & 3 & 7 & 18 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 6 & 42 & 60 \end{array} \right] \xrightarrow{R_4-6R_3} \\ &\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 0 & 1 & 3 & 7 & 18 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 6 & 6 \end{array} \right] \xrightarrow{\frac{1}{6}R_4} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 0 & 1 & 3 & 7 & 18 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{R_3-6R_4} \\ &\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 11 \\ 0 & 1 & 3 & 7 & 18 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{\substack{R_2-7R_4 \\ R_1-R_4}} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 10 \\ 0 & 1 & 3 & 0 & 11 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{\substack{R_1-R_3 \\ R_2-3R_3}} \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{R_1 - R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right]$$

Therefore, $\mathbf{x} = [5 \ 2 \ 3 \ 1]^T$ and the interpolating polynomial is

$$\boxed{p(t) = 5 + 2t + 3t^2 + t^3} \quad (1)$$

(b) **Determine the Lagrange polynomial interpolant to the same data and show that the resulting polynomial is equivalent to that obtained in part (a).**

The Lagrange form for the polynomial of degree three interpolating four points is

$$p(t) = y_1 \frac{(t - t_2)(t - t_3)(t - t_4)}{(t_1 - t_2)(t_1 - t_3)(t_1 - t_4)} + y_2 \frac{(t - t_1)(t - t_3)(t - t_4)}{(t_2 - t_1)(t_2 - t_3)(t_2 - t_4)} \\ + y_3 \frac{(t - t_1)(t - t_2)(t - t_4)}{(t_3 - t_1)(t_3 - t_2)(t_3 - t_4)} + y_4 \frac{(t - t_1)(t - t_2)(t - t_3)}{(t_4 - t_1)(t_4 - t_2)(t_4 - t_3)}$$

For the data points given, the equation above becomes

$$p(t) = 11 \frac{(t - 2)(t - 3)(t - 4)}{(1 - 2)(1 - 3)(1 - 4)} + 29 \frac{(t - 1)(t - 2)(t - 4)}{(2 - 1)(2 - 3)(2 - 4)} \\ + 65 \frac{(t - 1)(t - 2)(t - 3)}{(3 - 1)(3 - 2)(3 - 4)} + 125 \frac{(t - 1)(t - 2)(t - 3)}{(4 - 1)(4 - 2)(4 - 3)} \\ p(t) = \frac{11}{(-1)(-2)(-3)} (t^3 - 9t^2 + 26t - 24) + \frac{29}{(1)(-1)(-2)} (t^3 - 8t^2 + 19t - 12) \\ + \frac{65}{(2)(1)(-1)} (t^3 - 7t^2 + 14t - 8) + \frac{125}{(3)(2)(1)} (t^3 - 6t^2 + 11t - 6) \\ p(t) = \frac{-11}{6} (t^3 - 9t^2 + 26t - 24) + \frac{29}{2} (t^3 - 8t^2 + 19t - 12) \\ + \frac{-65}{2} (t^3 - 7t^2 + 14t - 8) + \frac{125}{6} (t^3 - 6t^2 + 11t - 6) \\ p(t) = \left(\frac{-11}{6} + \frac{29}{2} - \frac{65}{2} + \frac{125}{6} \right) t^3 + \left(-9 \cdot \frac{-11}{6} + -8 \cdot \frac{29}{2} + -7 \cdot \frac{-65}{2} + -6 \cdot \frac{125}{6} \right) t^2 \\ + \left(26 \cdot \frac{-11}{6} + 19 \cdot \frac{29}{2} + 14 \cdot \frac{-65}{2} + 11 \cdot \frac{125}{6} \right) t \\ + \left(-24 \cdot \frac{-11}{6} + -12 \cdot \frac{29}{2} + -8 \cdot \frac{-65}{2} + -6 \cdot \frac{125}{6} \right)$$

The Lagrange interpolating polynomial for the given data is

$$\boxed{p(t) = 5 + 2t + 3t^2 + t^3} \quad (2)$$

Note that this is also equivalent to the interpolant given in equation (1).

(c) **Compute the Newton polynomial interpolant to the same data using each of the three methods given in the text (triangular matrix, incremental interpolation and divided differences) and show that each produces the same result as the previous two methods.**

Solution.

(c.1) **Triangular Matrix.** With the Newton basis, we have the lower triangular linear system for interpolating four points:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & (t_2 - t_1) & 0 & 0 \\ 1 & (t_3 - t_1) & (t_3 - t_1)(t_3 - t_2) & 0 \\ 1 & (t_4 - t_1) & (t_4 - t_1)(t_4 - t_2) & (t_4 - t_1)(t_4 - t_2)(t_4 - t_3) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

For the data points given, the system becomes

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & (2 - 1) & 0 & 0 \\ 1 & (3 - 1) & (3 - 1)(3 - 2) & 0 \\ 1 & (4 - 1) & (4 - 1)(4 - 2) & (4 - 1)(4 - 2)(4 - 3) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 11 \\ 29 \\ 65 \\ 125 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 \\ 1 & 3 & 6 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 11 \\ 29 \\ 65 \\ 125 \end{bmatrix}$$

$$\mathbf{Ax} = \mathbf{y}$$

Since $\mathbf{A}$ is lower triangular, we can do forward substitution.

$$\begin{array}{lll} x_1 = 11 & & \\ x_1 + x_2 = 29 & 11 + x_2 = 29 & x_2 = 18 \\ x_1 + 2x_2 + 2x_3 = 65 & 2x_3 = 18 & x_3 = 9 \\ x_1 + 3x_2 + 6x_3 + 6x_4 = 125 & 6x_4 = 6 & x_4 = 1 \end{array}$$

The Newton interpolating polynomial is

$$\begin{aligned} p(t) &= x_1 + x_2(t - t_1) + x_3(t - t_1)(t - t_2) + x_4(t - t_1)(t - t_2)(t - t_3) \\ p(t) &= 11 + 18(t - 1) + 9(t - 1)(t - 2) + (t - 1)(t - 2)(t - 3) \\ p(t) &= 11 + 18t - 18 + 9t^2 - 27t + 18 + t^3 - 6t^2 + 11t - 6 \end{aligned}$$

$$\boxed{p(t) = 5 + 2t + 3t^2 + t^3} \quad (3)$$

Note that this is also equivalent to the interpolant given in equations (1) and (2).

(c.2) **Incremental Interpolation.** We begin with the first data point, $(t_1, y_1) = (1, 11)$ which is interpolated by the constant polynomial

$$p_1(t) = y_1 = 11$$

Incorporating the second data point, $(t_2, y_2) = (2, 29)$, we modify the previous polynomial so that it interpolates the new data point as well:

$$\begin{aligned} p_2(t) &= p_1(t) + \frac{y_2 - p_1(t_2)}{t_2 - t_1} (t - t_1) \\ p_2(t) &= 11 + \frac{29 - 11}{2 - 1} (t - 1) \\ p_2(t) &= 11 + 18(t - 1) \end{aligned}$$

Include the third data point, $(t_3, y_3) = (3, 65)$ we have

$$\begin{aligned}
p_3(t) &= p_2(t) + \frac{y_3 - p_2(t_3)}{(t_3 - t_1)(t_3 - t_2)} (t - t_1)(t - t_2) \\
p_3(t) &= 11 + 18(t - 1) + \frac{65 - 47}{(3 - 1)(3 - 2)} (t - 1)(t - 2) \\
p_3(t) &= 11 + 18t - 18 + \frac{18}{2} (t - 1)(t - 2) \\
p_3(t) &= 18t - 7 + 9t^2 - 27t + 18 \\
p_3(t) &= 9t^2 - 9t + 11
\end{aligned}$$

Lastly, include the fourth data point $(t_4, y_4) = (4, 125)$ we have

$$\begin{aligned}
p_4(t) &= p_3(t) + \frac{y_4 - p_3(t_4)}{(t_4 - t_1)(t_4 - t_2)(t_4 - t_3)} (t - t_1)(t - t_2)(t - t_3) \\
p_4(t) &= 9t^2 - 9t + 11 + \frac{125 - (9 \cdot 4^2 + 9 \cdot 4 - 11)}{(4 - 1)(4 - 2)(4 - 3)} (t - 1)(t - 2)(t - 3) \\
p_4(t) &= 9t^2 - 9t + 11 + \frac{6}{6} (t^3 - 6t^2 + 11t - 6) \\
p_4(t) &= 9t^2 - 9t + 11 + t^3 - 6t^2 + 11t - 6
\end{aligned}$$

$$\boxed{p(t) = 5 + 2t + 3t^2 + t^3} \tag{4}$$

Note that this is also equivalent to the interpolant given in equations (1), (2) and (3).

(c.3) **Newton Divided Differences.** The divided differences are defined using the recursive formula

$$\begin{aligned}
f[t_1, t_2, \dots, t_k] &= \frac{f[t_2, t_3, \dots, t_k] - f[t_1, t_2, \dots, t_{k-1}]}{t_k - t_1} \\
f[t_k] &= y_k \\
k &= 1, \dots, n.
\end{aligned}$$

This is done more efficiently using the table of divided differences:

$$\begin{array}{llll}
f[1] = 11 & & & \\
f[2] = 29 & f[1, 2] = \frac{29-11}{2-1} = 18 & & \\
f[3] = 65 & f[2, 3] = \frac{65-29}{3-2} = 36 & f[1, 2, 3] = \frac{36-18}{3-1} = 9 & \\
f[4] = 125 & f[3, 4] = \frac{125-65}{4-3} = 60 & f[2, 3, 4] = \frac{60-36}{4-2} = 12 & f[1, 2, 3, 4] = \frac{12-9}{4-1} = 1
\end{array}$$

The Newton polynomial is given by

$$\begin{aligned}
p(t) &= f[t_1] + f[t_1, t_2](t - t_1) + f[t_1, t_2, t_3](t - t_1)(t - t_2) + f[t_1, t_2, t_3, t_4](t - t_1)(t - t_1)(t - t_2)(t - t_3) \\
p(t) &= f[1] + f[1, 2](t - 1) + f[1, 2, 3](t - 1)(t - 2) + f[1, 2, 3, 4](t - 1)(t - 2)(t - 3) \\
p(t) &= 11 + 18t - 18 + 9t^2 - 27t + 18 + t^3 - 6t^2 + 11t - 6
\end{aligned}$$

$$\boxed{p(t) = 5 + 2t + 3t^2 + t^3} \tag{5}$$

In conclusion, monomial, Lagrange and Newton polynomial bases produce the same interpolant given the same set of data points.

1.2 Problem 7.12

(a) Verify directly that the first six Legendre polynomials $p_i(t)$ are indeed mutually orthogonal.

Solution. The Legendre polynomials are orthogonal over the interval $[-1, 1]$ with weight function $w(t) \equiv 1$. The first six Legendre polynomials are

$$\begin{aligned}p_0(t) &= 1 \\p_1(t) &= t \\p_2(t) &= \frac{3t^2 - 1}{2} \\p_3(t) &= \frac{5t^3 - 3t}{2} \\p_4(t) &= \frac{35t^4 - 30t^2 + 3}{8} \\p_5(t) &= \frac{63t^5 - 70t^3 + 15t}{8}\end{aligned}$$

Orthogonality implies that the inner product of any two Legendre polynomials,

$$\langle p_i, p_j \rangle = \int_{-1}^1 p_i(t) p_j(t) dt = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$\begin{aligned}\langle p_0, p_0 \rangle &= \int_{-1}^1 dt = t \Big|_{-1}^1 \\&= 1 - (-1) = \boxed{2} \\ \langle p_0, p_1 \rangle &= \int_{-1}^1 t dt = \frac{t^2}{2} \Big|_{-1}^1 \\&= \frac{1}{2} - \frac{(-1)^2}{2} = \boxed{0} \\ \langle p_0, p_2 \rangle &= \int_{-1}^1 \left(\frac{3}{2}t^2 - \frac{1}{2} \right) dt = \frac{1}{2}t^3 - \frac{t}{2} \Big|_{-1}^1 \\&= \left[\frac{1}{2}(1)^3 - \frac{1}{2} \right] - \left[\frac{1}{2}(-1)^3 - \frac{(-1)}{2} \right] = \boxed{0} \\ \langle p_0, p_3 \rangle &= \int_{-1}^1 \left(\frac{5}{2}t^3 - \frac{3}{2}t \right) dt = \frac{5}{8}t^4 - \frac{3}{4}t^2 \Big|_{-1}^1 \\&= \left[\frac{5}{8}(1)^4 - \frac{3}{4}(1)^2 \right] - \left[\frac{5}{8}(-1)^4 - \frac{3}{4}(-1)^2 \right] = \boxed{0} \\ \langle p_0, p_4 \rangle &= \int_{-1}^1 \left(\frac{35}{8}t^4 - \frac{30}{8}t^2 + \frac{3}{8} \right) dt = \frac{7}{8}t^5 - \frac{10}{8}t^3 + \frac{3}{8}t \Big|_{-1}^1 \\&= \left[\frac{7}{8}(1)^5 - \frac{10}{8}(1)^3 + \frac{3}{8}(1) \right] - \left[\frac{7}{8}(-1)^5 - \frac{10}{8}(-1)^3 + \frac{3}{8}(-1) \right] = \boxed{0}\end{aligned}$$

$$\begin{aligned}
\langle p_0, p_4 \rangle &= \int_{-1}^1 \left(\frac{35}{8}t^4 - \frac{30}{8}t^2 + \frac{3}{8} \right) dt = \frac{7}{8}t^5 - \frac{10}{8}t^3 + \frac{3}{8}t \Big|_{-1}^1 \\
&= \left[\frac{7}{8}(1)^5 - \frac{10}{8}(1)^3 + \frac{3}{8}(1) \right] - \left[\frac{7}{8}(-1)^5 - \frac{10}{8}(-1)^3 + \frac{3}{8}(-1) \right] = \boxed{0} \\
\langle p_0, p_5 \rangle &= \int_{-1}^1 \left(\frac{63}{8}t^5 - \frac{70}{8}t^3 + \frac{15}{8}t \right) dt = \frac{63}{48}t^6 - \frac{70}{32}t^4 + \frac{15}{16}t^2 \Big|_{-1}^1 \\
&= \left[\frac{63}{48}(1)^6 - \frac{70}{32}(1)^4 + \frac{15}{16}(1)^2 \right] - \left[\frac{63}{48}(-1)^6 - \frac{70}{32}(-1)^4 + \frac{15}{16}(-1)^2 \right] = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_1, p_1 \rangle &= \int_{-1}^1 t^2 dt = \frac{t^3}{3} \Big|_{-1}^1 \\
&= \frac{1^3}{3} - \frac{(-1)^3}{3} = \boxed{\frac{2}{3}} \\
\langle p_1, p_2 \rangle &= \int_{-1}^1 t \left(\frac{3}{2}t^3 - \frac{1}{2}t^2 \right) dt = \frac{3}{8}t^4 - \frac{1}{6}t^3 \Big|_{-1}^1 \\
&= \left[\frac{3}{8}(1)^4 - \frac{1}{6}(1)^3 \right] - \left[\frac{3}{8}(-1)^4 - \frac{1}{6}(-1)^3 \right] = \boxed{0} \\
\langle p_1, p_3 \rangle &= \int_{-1}^1 t \left(\frac{5}{2}t^3 - \frac{3}{2}t \right) dt = \frac{1}{2}t^5 - \frac{1}{2}t^3 \Big|_{-1}^1 \\
&= \left[\frac{1}{2}(1)^5 - \frac{1}{2}(1)^3 \right] - \left[\frac{1}{2}(-1)^5 - \frac{1}{2}(-1)^3 \right] = \boxed{0} \\
\langle p_1, p_4 \rangle &= \int_{-1}^1 t \left(\frac{35}{8}t^4 - \frac{30}{8}t^2 + \frac{3}{8} \right) dt = \frac{35}{48}t^6 - \frac{30}{24}t^4 + \frac{3}{16}t^2 \Big|_{-1}^1 \\
&= \left[\frac{35}{48}(1)^6 - \frac{30}{24}(1)^4 + \frac{3}{16}(1)^2 \right] - \left[\frac{35}{48}(-1)^6 - \frac{30}{24}(-1)^4 + \frac{3}{16}(-1)^2 \right] = \boxed{0} \\
\langle p_1, p_5 \rangle &= \int_{-1}^1 t \left(\frac{35}{8}t^4 - \frac{30}{8}t^2 - \frac{3}{8} \right) dt = \frac{35}{48}t^6 - \frac{30}{24}t^4 + \frac{3}{16}t^2 \Big|_{-1}^1 \\
&= \left[\frac{35}{48}(1)^6 - \frac{30}{24}(1)^4 + \frac{3}{16}(1)^2 \right] - \left[\frac{35}{48}(-1)^6 - \frac{30}{24}(-1)^4 + \frac{3}{16}(-1)^2 \right] = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_2, p_2 \rangle &= \int_{-1}^{-1} \left(\frac{3t^2 - 1}{2} \right)^2 dt = \int_{-1}^1 \left(\frac{9t^4 - 6t^2 + 1}{4} \right) dt = \frac{9}{20}t^5 - \frac{2}{4}t^3 + \frac{1}{4}t \Big|_{-1}^1 \\
&= \left[\frac{9}{20}(1)^5 - \frac{1}{2}(1)^3 + \frac{1}{4}(1) \right] - \left[\frac{9}{20}(-1)^5 - \frac{1}{2}(-1)^3 + \frac{1}{4}(-1) \right] = \boxed{\frac{2}{5}} \\
\langle p_2, p_3 \rangle &= \int_{-1}^1 \left(\frac{3t^2 - 1}{2} \right) \left(\frac{5t^3 - 3t}{2} \right) dt = \frac{1}{4} \int_{-1}^1 (15t^5 - 9t^3 - 5t^3 + 3t) dt = \frac{1}{4} \left\{ \frac{15}{6}t^6 - \frac{14}{4}t^4 + \frac{3}{2}t^2 \right\} \Big|_{-1}^1 \\
&= \frac{1}{4} \left\{ \left[\frac{15}{6}(1)^6 - \frac{14}{4}(1)^4 + \frac{3}{2}(1)^2 \right] - \left[\frac{15}{6}(-1)^6 - \frac{14}{4}(-1)^4 + \frac{3}{2}(-1)^2 \right] \right\} = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_2, p_4 \rangle &= \int_{-1}^1 \left(\frac{3t^2 - 1}{2} \right) \left(\frac{35t^4 - 30t^2 + 3}{8} \right) dt \\
&= \frac{1}{16} \int_{-1}^1 (189t^7 - 210t^5 + 45t^3 - 63t^5 + 70t^3 - 15t) dt \\
&= \frac{1}{16} \left\{ \frac{189}{8}t^8 - \frac{273}{6}t^6 + \frac{115}{4}t^4 - \frac{15}{2}t^2 \right\}_{-1}^1 \\
&= \frac{1}{16} \left\{ \left[\frac{189}{8}(1)^8 - \frac{273}{6}(1)^6 + \frac{115}{4}(1)^4 - \frac{15}{2}(1)^2 \right] \right\} \\
&\quad - \frac{1}{16} \left\{ \left[\frac{189}{8}(-1)^8 - \frac{273}{6}(-1)^6 + \frac{115}{4}(-1)^4 - \frac{15}{2}(-1)^2 \right] \right\} = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_3, p_3 \rangle &= \int_{-1}^1 \left(\frac{5t^3 - 3t}{2} \right)^2 dt = \int_{-1}^1 \left(\frac{25t^6 - 30t^4 + 9t^2}{4} \right) dt = \frac{25}{28}t^7 - \frac{6}{4}t^5 + \frac{3}{4}t^3 \Big|_{-1}^1 \\
&= \left[\frac{25}{28}(1)^7 - \frac{6}{4}(1)^5 + \frac{3}{4}(1)^3 \right] - \left[\frac{25}{28}(-1)^7 - \frac{6}{4}(-1)^5 + \frac{3}{4}(-1)^3 \right] = \boxed{\frac{2}{7}}
\end{aligned}$$

$$\begin{aligned}
\langle p_3, p_4 \rangle &= \int_{-1}^1 \left(\frac{5t^3 - 3t}{2} \right) \left(\frac{35t^4 - 30t^2 + 3}{8} \right) dt \\
&= \frac{1}{16} \int_{-1}^1 (175t^7 - 150t^5 + 15t^3 - 75t^5 + 90t^3 - 9t) dt \\
&= \frac{1}{16} \left\{ \frac{175}{8}t^8 - \frac{225}{6}t^6 + \frac{105}{4}t^4 - \frac{9}{2}t^2 \right\}_{-1}^1 \\
&= \frac{1}{16} \left\{ \left[\frac{175}{8}(1)^8 - \frac{225}{6}(1)^6 + \frac{105}{4}(1)^4 - \frac{9}{2}(1)^2 \right] \right\} \\
&\quad - \left\{ \left[\frac{175}{8}(-1)^8 - \frac{225}{6}(-1)^6 + \frac{105}{4}(-1)^4 - \frac{9}{2}(-1)^2 \right] \right\} = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_3, p_5 \rangle &= \int_{-1}^1 \left(\frac{5t^3 - 3t}{2} \right) \left(\frac{63t^5 - 70t^3 + 15t}{8} \right) dt \\
&= \frac{1}{16} \int_{-1}^1 (315t^8 - 350t^6 + 75t^4 - 189t^6 + 210t^4 - 45t^2) dt \\
&= \frac{1}{16} \{ 35t^9 - 77t^7 + 57t^5 - 15t^3 \}_{-1}^1 \\
&= \frac{1}{16} \{ [35(1)^9 - 77(1)^7 + 57(1)^5 - 15(1)^3] \} \\
&\quad - \frac{1}{16} \{ [35(-1)^9 - 77(-1)^7 + 57(-1)^5 - 15(-1)^3] \} = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_4, p_4 \rangle &= \int_{-1}^1 \left(\frac{35t^4 - 30t^2 - 3}{8} \right)^2 dt = \frac{1}{64} \int_{-1}^1 (1225t^8 - 2100t^6 + 1110t^4 - 180t^2 + 9) dt \\
&= \frac{1}{64} \left\{ \frac{1225}{9}t^9 - 300t^7 + 222t^5 - 60t^3 + 9t \right\}_{-1}^1 \\
&= \frac{1}{64} \left\{ \left[\frac{1225}{9}(1)^9 - 300(1)^7 + 222(1)^5 - 60(1)^3 + 9(1) \right] \right\} \\
&\quad - \frac{1}{64} \left\{ \left[\frac{1225}{9}(-1)^9 - 300(-1)^7 + 222(-1)^5 - 60(-1)^3 + 9(-1) \right] \right\} = \boxed{\frac{2}{9}} \\
\langle p_4, p_5 \rangle &= \int_{-1}^1 \left(\frac{35t^4 - 30t^2 - 3}{8} \right) \left(\frac{63t^5 - 70t^3 + 15t}{8} \right) dt \\
&= \frac{1}{64} \int_{-1}^1 (2205t^9 - 4340t^7 + 2436t^5 - 240t^3 - 45t) dt \\
&= \left\{ \frac{441}{2}t^{10} - \frac{1085}{2}t^8 + 406t^6 - 60t^4 - \frac{45}{2}t^2 \right\}_{-1}^1 \\
&= \left\{ \left[\frac{441}{2}(1)^{10} - \frac{1085}{2}(1)^8 + 406(1)^6 - 60(1)^4 - \frac{45}{2}(1)^2 \right] \right\} \\
&= \left\{ \left[\frac{441}{2}(1)^{10} - \frac{1085}{2}(-1)^8 + 406(-1)^6 - 60(-1)^4 - \frac{45}{2}(-1)^2 \right] \right\} = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
\langle p_5, p_5 \rangle &= \int_{-1}^1 \left(\frac{63t^5 - 70t^3 + 15t}{8} \right)^2 dt \\
&= \frac{1}{64} \int_{-1}^1 (3969t^{10} + 4900t^6 + 225t^2 - 8820t^8 - 2100t^4 + 1890t^6) dt \\
&= \frac{1}{64} \left\{ \frac{3969}{11}t^{11} + 430t^7 + 75t^3 - 980t^9 - 420t^5 \right\}_{-1}^1 \\
&= \left\{ \frac{3969}{11}(1)^{11} + 430(1)^7 + 75(1)^3 - 980(1)^9 - 420(1)^5 \right\} \\
&\quad - \left\{ \frac{3969}{11}(-1)^{11} + 430(-1)^7 + 75(-1)^3 - 980(-1)^9 - 420(-1)^5 \right\} = \boxed{\frac{2}{11}}
\end{aligned}$$

In general, the Legendre polynomials obey the orthogonality relationship

$$\int_{-1}^1 p_n(t)p_m(t) dt = \frac{2}{2n+1}\delta_{mn}, \quad \delta_{mn} = \begin{cases} 1, m = n \\ 0, m \neq n \end{cases} \quad (6)$$

where δ_{mn} is the Kronecker delta function.

(b) Verify directly that they satisfy the three-term recurrence relation

$$(k+1)p_{k+1}(t) = (2k+1)t p_k(t) - k p_{k-1}(t) \quad (7)$$

Solution. Let's start with initial values $k=0$ and $p_0(t)=1$ to obtain $p_1(t)$.

$$(0+1)p_1(t) = (2 \cdot 0 + 1)t p_0(t) - 0p_{-1}(t)$$

$$\boxed{p_1(t) = t}$$

Next, obtain $p_2(t)$ at $k = 1$:

$$(1 + 1) p_2(t) = (2 \cdot 1 + 1) t p_1(t) - p_0(t)$$

$$2 p_2(t) = 3t^2 - 1$$

$$\boxed{p_2(t) = \frac{3t^2 - 1}{2}}$$

Next, obtain $p_3(t)$ at $k = 2$:

$$(2 + 1) p_3(t) = (2 \cdot 2 + 1) t p_2(t) - 2 p_1(t)$$

$$3 p_3(t) = 5t \left(\frac{3t^2 - 1}{2} \right) - 2t = \frac{15t^3}{2} - \frac{9t}{2}$$

$$\boxed{p_3(t) = \frac{5t^3 - 3t}{2}}$$

Next, obtain $p_4(t)$ using $k = 3$:

$$(3 + 1) p_4(t) = (2 \cdot 3 + 1) t p_3(t) - 3 p_2(t)$$

$$4 p_4(t) = 7t \left(\frac{5t^3 - 3t}{2} \right) - 3 \left(\frac{3t^2 - 1}{2} \right)$$

$$4 p_4(t) = \frac{35t^4 - 30t^2 + 3}{2}$$

$$\boxed{p_4(t) = \frac{35t^4 - 30t^2 + 3}{3}}$$

Lastly, get $p_5(t)$ using $k = 4$:

$$(4 + 1) p_5(t) = (2 \cdot 4 + 1) p_4(t) - 4 p_3(t)$$

$$5 p_5(t) = 9t \left(\frac{5t^3 - 3t}{2} \right) - 4 \left(\frac{5t^3 - 3t}{2} \right)$$

$$5 p_5(t) = \frac{315t^5 - 270t^3 - 27t}{8} - 10t^3 + 6t$$

$$5 p_5(t) = \frac{315t^5 - 350t^3 + 75t}{8}$$

$$\boxed{p_5(t) = \frac{63t^5 - 70t^3 + 15t}{8}}$$

(c) Express each of the first six monomials m_k as a linear combination of the first six Legendre polynomials p_k .

Solution. We must solve constants c_i such that

$$m_k = \sum_{i=0}^5 c_i p_i \quad \forall k = 0, \dots, 5. \quad (8)$$

This approach is analogous to interpolating monomials using Legendre polynomials as basis functions.

First, we interpolate m_0 . We have a trivial solution

$$m_0 = 1 = p_0$$

$$\boxed{m_0 = p_0}$$

Next, we interpolate m_1 . We also have a trivial solution

$$m_1 = t = 0p_0 + 1p_1$$

$m_1 = p_1$

Next, interpolate m_2 . We now have an equation

$$\begin{aligned} m_2 = t^2 &= c_0p_0 + c_1p_1 + c_2p_2 \\ &= c_0 + c_1t + c^2 \left(\frac{3t^2 - 1}{2} \right) \\ &= c_0 + c_1t + \frac{3}{2}c_2t^2 - \frac{c_2}{2} \end{aligned}$$

which yields a system of equations

$$\begin{aligned} c_0 - \frac{c_2}{2} &= 0 \\ c_1 &= 0 \\ \frac{3}{2}c_2 &= 1 \end{aligned}$$

with solutions

$$c_0 = \frac{1}{3}, c_1 = 0, c_2 = \frac{2}{3}$$

Therefore, we have the relationship

$m_2 = \frac{1}{3}p_0 + \frac{2}{3}p_2$

Next, interpolate m_3 . We now have an equation

$$\begin{aligned} m_3 = t^3 &= c_0p_0 + c_1p_1 + c_2p_2 + c_3p_3 \\ &= c_0 + c_1t + c_2 \left(\frac{3t^2 - 1}{2} \right) + c_3 \left(\frac{5t^3 - 3t}{2} \right) \\ &= c_0 + c_1t + \frac{3}{2}c_2t^2 - \frac{c_2}{2} + \frac{5}{2}c_3t^3 - \frac{3}{2}c_3t \end{aligned}$$

which yields a system of equations

$$\begin{aligned} c_0 - \frac{c_2}{2} &= 0 \\ c_1 - \frac{3}{2}c_3 &= 0 \\ \frac{3}{2}c_2 &= 0 \\ \frac{5}{2}c_3 &= 1 \end{aligned}$$

with solutions

$$c_0 = 0, c_1 = \frac{3}{5}, c_2 = 0, c_3 = \frac{2}{5}$$

Therefore, we have the relationship

$$m_3 = \frac{3}{5}p_1 + \frac{2}{5}p_3$$

Next, interpolate m_4 . We now have an equation

$$\begin{aligned} m_4 = t^4 &= c_0p_0 + c_1p_1 + c_2p_2 + c_3p_3 + c_4p_4 \\ &= c_0 + c_1t + c_2\left(\frac{3t^2-1}{2}\right) + c_3\left(\frac{5t^3-3t}{2}\right) + c_4\left(\frac{35t^4-30t^2+3}{8}\right) \\ &= c_0 + c_1t + \frac{3}{2}c_2t^2 - \frac{c_2}{2} + \frac{5}{2}c_3t^3 - \frac{3}{2}c_3t + \frac{35}{8}c_4t^4 - \frac{30}{8}c_4t^2 + \frac{3}{8}c_4 \end{aligned}$$

which yields a system of equations

$$\begin{aligned} c_0 - \frac{c_2}{2} + \frac{3}{8}c_4 &= 0 \\ c_1 - \frac{3}{2}c_3 &= 0 \\ \frac{3}{2}c_2 - \frac{30}{8}c_4 &= 0 \\ \frac{5}{2}c_3 &= 0 \\ \frac{35}{8}c_4 &= 1 \end{aligned}$$

with solutions

$$c_0 = \frac{1}{5}, c_1 = 0, c_2 = \frac{4}{7}, c_3 = 0, c_4 = \frac{8}{35}$$

Therefore, we have the relationship

$$m_4 = \frac{1}{5}p_0 + \frac{4}{7}p_2 + \frac{8}{35}p_4$$

Lastly, interpolate m_5 . We now have an equation

$$\begin{aligned} t^5 &= c_0p_0 + c_1p_1 + c_2p_2 + c_3p_3 + c_4p_4 + c_5p_5 \\ &= c_0 + c_1t + c_2\left(\frac{3t^2-1}{2}\right) + c_3\left(\frac{5t^3-3t}{2}\right) + c_4\left(\frac{35t^4-30t^2+3}{8}\right) + c_5\left(\frac{63t^5-70t^3+15t}{8}\right) \\ &= c_0 + c_1t + \frac{3}{2}c_2t^2 - \frac{c_2}{2} + \frac{5}{2}c_3t^3 - \frac{3}{2}c_3t + \frac{35}{8}c_4t^4 - \frac{30}{8}c_4t^2 + \frac{3}{8}c_4 + \frac{63}{8}c_5t^5 - \frac{70}{8}c_5t^3 + \frac{15}{8}c_5t \end{aligned}$$

which yields a system of equations

$$\begin{aligned} c_0 - \frac{c_2}{2} + \frac{3}{8}c_4 &= 0 \\ c_1 - \frac{3}{2}c_3 + \frac{15}{8}c_5 &= 0 \\ \frac{3}{2}c_2 - \frac{30}{8}c_4 &= 0 \\ \frac{5}{2}c_3 - \frac{70}{8}c_5 &= 0 \\ \frac{35}{8}c_4 &= 0 \\ \frac{63}{8}c_5 &= 1 \end{aligned}$$

with solutions

$$c_0 = 0, c_1 = \frac{27}{63}, c_2 = 0, c_3 = \frac{4}{9}, c_4 = 0, c_5 = \frac{8}{63}$$

Therefore, we have the relationship

$$m_5 = \frac{27}{63}p_1 + \frac{4}{9}p_3 + \frac{8}{63}p_5$$

Further Notes. In general, interpolating a polynomial from a monomial basis using Legendre polynomials as basis functions, it is required to solve the system

$$\begin{bmatrix} \Delta_{00} & \Delta_{01} & \cdots & \Delta_{0n} \\ \Delta_{10} & \Delta_{11} & \cdots & \Delta_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \Delta_{n0} & \Delta_{n1} & \cdots & \Delta_{nn} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Where Δ_{ij} is the coefficient of the term with i th degree in the j th Legendre polynomial, c_i 's are the constants we want to solve and x_i is the coefficient of the term with i th degree in the polynomial of monomial basis that we want to interpolate.

Further note that $\Delta_{ij} = 0$ for $i > j$ since there are no terms with i th degree in the j th Legendre polynomial. For example, there is no term with degree 3 in $p_2(t)$, hence $\Delta_{32} = 0$. With this simplification, the resulting system will now have a upper triangular matrix.

$$\begin{bmatrix} \Delta_{00} & \Delta_{01} & \cdots & \Delta_{0n} \\ 0 & \Delta_{11} & \cdots & \Delta_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Delta_{nn} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$\Delta \mathbf{c} = \mathbf{x}$$

Lastly, if the vector $\mathbf{x}$ is the natural basis vector $[0 \ 0 \ \cdots \ 1]$ it can be shown that

$$\sum_{i=0}^n c_i = 1 \tag{9}$$

Proof: We know that $m_k(1) = p_k(t) = 1, k = 0, \dots$. Using equation (8) and $t = 1, n = 5$, we have

$$1 = \sum_{i=0}^5 c_i p_i = \sum_{i=0}^5 c_i$$

Let us have an example. Interpolating $m_5(t) = t^5$ using Legendre polynomials, we have

$$\begin{bmatrix} 1 & 0 & \frac{-1}{2} & 0 & \frac{3}{8} & 0 \\ 0 & 1 & 0 & \frac{-3}{2} & 0 & \frac{15}{8} \\ 0 & 0 & \frac{3}{2} & 0 & \frac{-30}{8} & 0 \\ 0 & 0 & 0 & \frac{5}{2} & 0 & \frac{-70}{8} \\ 0 & 0 & 0 & 0 & \frac{35}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{63}{8} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Solving for c_i 's can be executed using backward substitution. This is already done earlier.

$$\mathbf{c} = [0 \ \frac{27}{63} \ 0 \ \frac{4}{9} \ 0 \ \frac{8}{63}], \quad \sum_{i=0}^n c_i = \frac{27}{63} + \frac{4}{9} + \frac{8}{63} = 1$$

Thus, proving the claim in equation (9). $\square$

2 Numerical Integration and Differentiation

2.1 Problem 8.1

(a) Compute the approximate value of $\int_0^1 x^3 dx$, first by the midpoint rule and then by the trapezoid rule.

Solution. With the midpoint rule,

$$M = (b - a) f\left(\frac{a + b}{2}\right)$$

$$M = (1 - 0) \left(\frac{1}{2}\right)^3$$

$$\boxed{M = \frac{1}{8}}$$

With the trapezoid rule, we have

$$T = \frac{b - a}{2} [f(a) + f(b)]$$

$$T = \left(\frac{1 - 0}{2}\right) [1^3 + 0^3]$$

$$\boxed{T = \frac{1}{2}}$$

(b) Use the difference between these two results to estimate the error in each of them.

Solution. The error in computation is given by

$$E \approx \frac{T - M}{3}$$

$$E \approx \frac{\frac{1}{2} - \frac{1}{8}}{3}$$

$$\boxed{E \approx \frac{1}{8}}$$

The error in M is approximately 1/8 while error in T is approximately -1/4. This follows from the fact the error in T is twice the error in M.

(c) Combine the two results to obtain the Simpson's rule approximation to the integral.

Solution. Simpson's rule is obtained by

$$S = \frac{2}{3}M + \frac{1}{3}T$$

$$S = \frac{2}{3} \left(\frac{1}{8}\right) + \frac{1}{3} \left(\frac{1}{2}\right) = \frac{1}{12} + \frac{1}{6}$$

$$\boxed{S = \frac{1}{4}}$$

(d) **Would you expect the latter to be exact for this problem? Why?**

Solution. The integral is actually

$$I = \int_0^1 x^3 dx = \frac{x^4}{4} \Big|_0^1 = \boxed{\frac{1}{4}}$$

And $I = S$. The error of Simpson's method depends of the fourth and higher derivatives of the integrand that will vanish in cubic polynomials. We expect Simpson's rule to be exact in this problem.

2.2 Problem 8.2

(a) **Using the composite midpoint quadrature rule, compute the approximate value for the integral $\int_0^1 x^3 dx$, using a mesh size (subinterval length) of $h = 0.5$ and also using a mesh size of $h = 1$.**

Solution. Using mesh size of $h = 1$,

$$M(h) = (1 - 0) \left(\frac{1}{2}\right)^3 = \frac{1}{8} = \boxed{0.125}$$

Using mesh size of $h = 0.5$,

$$M\left(\frac{h}{2}\right) = (0.5 - 0)(0.25)^3 + (1 - 0.5)(0.75)^3 = \boxed{0.21875}$$

(b) **Based on the two approximate values computed in part (a), use Richardson extrapolation to compute a more accurate approximation to the integral.**

Solution. In this particular example, let $p = 2$, which is the order of accuracy of the midpoint quadrature rule. The extrapolated value is given by

$$\begin{aligned} a_0 &= M(h) + \frac{M(h) - M(h/2)}{2^{-2} - 1} \\ &= 0.125 + \frac{0.125 - 0.21875}{0.25 - 1} \\ &= 0.125 + \frac{-0.09375}{-0.75} \\ &= \boxed{0.25} \end{aligned}$$

(c) **Would you expect the extrapolated result computed in part (b) to be exact in this case? Why?**

Solution. In this particular extrapolation, $p = 2$ and $r = 4$ which is based on the error analysis of the midpoint rule. The accuracy of the extrapolation will be greater than the accuracy of the midpoint method. In this case, it is $r = 4$. This really gives exact results in cubic polynomials as integrands.

2.3 Problem 8.4

Fill in the details of the derivation of the error estimates for the midpoint and trapezoid quadrature rules given in Section 8.3.1. In particular, show that the odd-order terms drop out in both cases, as claimed.

Solution. The error in the midpoint rule can be estimated using a Taylor series expansion about the midpoint $m = (b + a)/2$ of the interval $[a, b]$

$$f(x) = f(m) + f'(m)(x - m) + f''(m)\frac{(x - m)^2}{2} + f^{(3)}(m)\frac{(x - m)^3}{6} \quad (10)$$

$$+ f^{(4)}(m)\frac{(x - m)^4}{24} + f^{(5)}(m)\frac{(x - m)^5}{120} + f^{(6)}(m)\frac{(x - m)^6}{720} + \dots \quad (11)$$

We let $I = \int_b^a f(x) dx$. Integrating both sides from a to b :

$$I(f) = f(m)(b - a) + f'(m)\frac{(x - m)^2}{2}\Big|_a^b + f''(m)\frac{(x - m)^3}{6}\Big|_a^b + f^{(3)}(m)\frac{(x - m)^4}{24}\Big|_a^b \quad (12)$$

$$+ f^{(4)}(m)\frac{(x - m)^5}{120}\Big|_a^b + f^{(5)}(m)\frac{(x - m)^6}{720}\Big|_a^b + f^{(6)}(m)\frac{(x - m)^7}{5040}\Big|_a^b + \dots \quad (13)$$

Consider evaluating

$$\begin{aligned} (x - m)^n \Big|_a^b &= (b - m)^n - (a - m)^n \\ &= \left(b - \frac{a + b}{2}\right)^n - \left(a - \frac{a + b}{2}\right)^n \\ &= \left(\frac{b - a}{2}\right)^n - \left(\frac{a - b}{2}\right)^n \\ &= \frac{1}{2^n} [(b - a)^n - (-1)^n (b - a)^n] \end{aligned}$$

Looking at possible values of $n > 0$, it can be shown that

$$(x - m)^n \Big|_a^b = \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{(b - a)^n}{2^{n-1}} & \text{if } n \text{ is odd} \end{cases} \quad (14)$$

Applying equation (14) to equation (12),

$$\begin{aligned} I(f) &= f(m)(b - a) + \frac{f'(m)}{2} \cdot 0 + \frac{f''(m)}{6} \frac{(x - m)^3}{2^2} + \frac{f^{(3)}(m)}{24} \cdot 0 \\ &\quad + \frac{f^{(4)}(m)}{120} \frac{(x - m)^5}{2^4} + \frac{f^{(5)}(m)}{720} \cdot 0 + \frac{f^{(6)}(m)}{5040} \frac{(b - a)^7}{2^6} + \dots \end{aligned}$$

Notice that the odd-order terms will drop out, yielding

$$I(f) = f(m)(b - a) + \frac{f''(m)}{24}(b - a)^3 + \frac{f^{(4)}(m)}{1920}(b - a)^5 + \dots \quad (15)$$

$$\boxed{I(f) = M(f) + E(f) + F(f)} \quad (16)$$

Where E and F represent the first two terms in the error expansion for the midpoint rule. To derive a comparable error expansion in the trapezoid quadrature rule, we substitute $x = a$ and $x = b$ into the Taylor series.

$$\begin{aligned} f(a) &= f(m) + f'(m)(a-m) + f''(m)\frac{(a-m)^2}{2} + f^{(3)}(m)\frac{(a-m)^3}{6} \\ &\quad + f^{(4)}(m)\frac{(a-m)^4}{24} + f^{(5)}(m)\frac{(a-m)^5}{120} + f^{(6)}(m)\frac{(a-m)^6}{720} + \dots \\ f(b) &= f(m) + f'(m)(b-m) + f''(m)\frac{(b-m)^2}{2} + f^{(3)}(m)\frac{(b-m)^3}{6} \\ &\quad + f^{(4)}(m)\frac{(b-m)^4}{24} + f^{(5)}(m)\frac{(b-m)^5}{120} + f^{(6)}(m)\frac{(b-m)^6}{720} + \dots \end{aligned}$$

Add the two series together, yielding

$$f(a) + f(b) = 2f(m) + f'(m)[(a-m) + (b-m)] + \frac{f''(m)}{2}[(a-m)^2 + (b-m)^2] \quad (17)$$

$$+ \frac{f^{(3)}(m)}{6}[(a-m)^3 + (b-m)^3] + \frac{f^{(4)}(m)}{24}[(a-m)^4 + (b-m)^4] \quad (18)$$

$$+ \frac{f^{(5)}(m)}{120}[(a-m)^5 + (b-m)^5] + \frac{f^{(6)}(m)}{720}[(a-m)^6 + (b-m)^6] \quad (19)$$

Now, consider evaluating

$$\begin{aligned} (a-m)^n + (b-m)^n &= \left(a - \frac{b+a}{2}\right)^n + \left(b - \frac{b+a}{2}\right)^n \\ &= \left(\frac{a-b}{2}\right)^n + \left(\frac{b-a}{2}\right)^n \\ &= \frac{1}{2^n}[(a-b)^n + (b-a)^n] \\ &= \frac{1}{2^n}[(a-b)^n + (-1)^n(a-b)^n] \end{aligned}$$

Looking at possible values of $n > 0$, it can be shown that

$$(a-m)^n + (b-m)^n = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \frac{(a-b)^n}{2^{n-1}} = \frac{(b-a)^n}{2^{n-1}} & \text{if } n \text{ is even} \end{cases} \quad (20)$$

Applying equation (20) to equation (17),

$$\begin{aligned} f(a) + f(b) &= 2f(m) + f'(m) \cdot 0 + \frac{f''(m)}{2} \frac{(b-a)^2}{2} + \frac{f^{(3)}(m)}{6} \cdot 0 \\ &\quad + \frac{f^{(4)}(m)}{24} \frac{(b-a)^4}{2^3} + \frac{f^{(5)}(m)}{120} \cdot 0 + \frac{f^{(6)}(m)}{720} [(a-m)^6 + (b-m)^6] + \dots \end{aligned}$$

Notice again that the odd-order terms will drop out, yielding

$$\begin{aligned} f(a) + f(b) &= 2f(m) + \frac{f''(m)}{4}(b-a)^2 + \frac{f^{(4)}(m)}{192}(b-a)^4 + \dots \\ f(m) &= \frac{f(a) + f(b)}{2} - \left(\frac{f''(m)}{8}(b-a)^2 + \frac{f^{(4)}(m)}{384}(b-a)^4 + \dots \right) \end{aligned}$$

$$f(m) = T(f) - \left(\frac{f''(m)}{8}(b-a)^2 + \frac{f^{(4)}(m)}{384}(b-a)^4 + \dots \right)$$

Substituting $f(m)$ in the midpoint expansion given in equation (15),

$$I(f) = \left\{ T(f) - \left(\frac{f''(m)}{8}(b-a)^2 + \frac{f^{(4)}(m)}{384}(b-a)^4 + \dots \right) \right\} (b-a) + \frac{f''(m)}{24}(b-a)^3 + \frac{f^{(4)}(m)}{1920}(b-a)^5 + \dots$$

$$I(f) = T(f)(b-a) + \left(\frac{-1}{8} + \frac{1}{24} \right) \cdot f''(m)(b-a)^3 + \left(\frac{-1}{384} + \frac{1}{1920} \right) \cdot f^{(4)}(m)(b-a)^5 + \dots$$

$$I(f) = T(f)(b-a) - \frac{1}{12}f''(m)(b-a)^3 - \frac{1}{480}f^{(4)}(m)(b-a)^5 + \dots$$

$$\boxed{I(f) = T(f) - 2E(f) - 4F(f) - \dots}$$

Note that

$$T(f) - M(f) = T(f) - \left\{ I(f) - \left(\frac{f''(m)}{24}(b-a)^3 + \frac{f^{(4)}(m)}{1920}(b-a)^5 + \dots \right) \right\}$$

$$T(f) - M(f) = T(f) - \{ (T(f) - 2E(f) - 4F(f) - \dots) - (E(f) + F(f) + \dots) \}$$

$$\boxed{T(f) - M(f) = 3E(f) + 5F(f) + \dots}$$

and hence the difference between the two quadrature rules provides an estimate for the dominant term in their error expansions,

$$\boxed{E(f) \approx \frac{T(f) - M(f)}{3}} \quad (21)$$

provided that the length of the interval is sufficiently small that $(b-a)^5 \ll (b-a)^3$, and the integrand f is such that $f^{(4)}$ is well-behaved. $\square$

2.4 Problem 8.6

Suppose that Lagrange interpolation at a given set of nodes $x_1, \dots, x_n$ is used to derive a quadrature rule. Prove that the corresponding weights are given by the integrals of the Lagrange basis functions, $w_i = \int_a^b \ell_i(x) dx$, $i = 1, \dots, n$.

Solution. Using the method of undetermined coefficients, we choose the weights so that the rule integrates the first n polynomial basis functions exactly, which results in a system of n equations in n unknowns with the Lagrange basis, for example, this method results in the system of moment equations

$$\begin{aligned} w_1 y_1 \ell_1(t_1) + w_2 y_1 \ell_1(t_2) + \dots + w_n y_1 \ell_1(t_n) &= \int_a^b y_1 \ell_1(t) dt = y_1 \int_a^b \ell_1(t) dt \\ w_1 y_2 \ell_2(t_1) + w_2 y_2 \ell_2(t_2) + \dots + w_n y_2 \ell_2(t_n) &= \int_a^b y_2 \ell_2(t) dt = y_2 \int_a^b \ell_2(t) dt \\ &\vdots \\ w_1 y_n \ell_n(t_1) + w_2 y_n \ell_n(t_2) + \dots + w_n y_n \ell_n(t_n) &= \int_a^b y_n \ell_n(t) dt = y_n \int_a^b \ell_n(t) dt \end{aligned}$$

Writing this system in matrix form, we have

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}^T \begin{bmatrix} \ell_1(t_1) & \ell_1(t_2) & \dots & \ell_1(t_n) \\ \ell_2(t_1) & \ell_2(t_2) & \dots & \ell_2(t_n) \\ \vdots & \vdots & \ddots & \vdots \\ \ell_n(t_1) & \ell_n(t_2) & \dots & \ell_n(t_n) \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}^T \begin{bmatrix} \int_a^b \ell_1(t) dt \\ \int_a^b \ell_2(t) dt \\ \vdots \\ \int_a^b \ell_n(t) dt \end{bmatrix}$$

Multiply vector $\left[\frac{1}{y_1} \frac{1}{y_2} \cdots \frac{1}{y_n}\right]$ to both sides of the equation, yielding

$$\begin{bmatrix} \ell_1(t_1) & \ell_1(t_2) & \cdots & \ell_1(t_n) \\ \ell_2(t_1) & \ell_2(t_2) & \cdots & \ell_2(t_n) \\ \vdots & \vdots & \ddots & \vdots \\ \ell_n(t_1) & \ell_n(t_2) & \cdots & \ell_n(t_n) \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} \int_a^b \ell_1(t) \, dt \\ \int_a^b \ell_2(t) \, dt \\ \vdots \\ \int_a^b \ell_n(t) \, dt \end{bmatrix}$$

$$\mathbf{L}\mathbf{w} = \mathbf{i}$$

Since the property of Lagrange basis states that

$$\ell_j(t_i) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}, \quad i = 1, \dots, n$$

The square matrix $\mathbf{L}$ is just an identity matrix, hence we have a solution

$$\mathbf{w} = \mathbf{i}$$

$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} \int_a^b \ell_1(t) \, dt \\ \int_a^b \ell_2(t) \, dt \\ \vdots \\ \int_a^b \ell_n(t) \, dt \end{bmatrix}$$

$$w_i = \int_a^b \ell_i(x) \, dx, \quad i = 1, \dots, n.$$

Which are the corresponding weights, hence proving the claim. $\square$

3 Initial Value Problems for ODEs

3.1 Problem 9.2

Write each of the following ODEs as an equivalent first-order system of ODEs:

(a) Van der Pol Equation:

$$y'' = y' (1 - y^2) - y$$

Solution. Define a new vector $\mathbf{U}$ such that

$$\begin{aligned}\mathbf{U} &= \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} y \\ y' \end{bmatrix} \\ \mathbf{U}' &= \begin{bmatrix} u'_1 \\ u'_2 \end{bmatrix} = \begin{bmatrix} y' \\ y'' \end{bmatrix}\end{aligned}$$

So that we can have

$$\begin{bmatrix} u'_1 \\ u'_2 \end{bmatrix} = \begin{bmatrix} u_2 \\ u_2 (1 - u_1^2) - u_1 \end{bmatrix}$$

(b) Blasius equation:

$$y''' = -yy''$$

Solution. Define a new vector $\mathbf{U}$ such that

$$\begin{aligned}\mathbf{U} &= \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} \\ \mathbf{U}' &= \begin{bmatrix} u'_1 \\ u'_2 \\ u'_3 \end{bmatrix} = \begin{bmatrix} y' \\ y'' \\ y''' \end{bmatrix}\end{aligned}$$

So that we can have

$$\begin{bmatrix} u'_1 \\ u'_2 \\ u'_3 \end{bmatrix} = \begin{bmatrix} u_2 \\ u_3 \\ -u_1 u_3 \end{bmatrix}$$

(c) Newton's Second Law of Motion for two-body problem:

$$\begin{aligned}y''_1 &= -GM y_1 / (y_1^2 + y_2^2)^{3/2} \\ y''_2 &= -GM y_2 / (y_1^2 + y_2^2)^{3/2}\end{aligned}$$

Solution. Define a new vector $\mathbf{U}$ such that

$$\mathbf{U} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_1' \\ y_2 \\ y_2' \end{bmatrix}$$

$$\mathbf{U}' = \begin{bmatrix} u_1' \\ u_2' \\ u_3' \\ u_4' \end{bmatrix} = \begin{bmatrix} y_1' \\ y_1'' \\ y_2' \\ y_2'' \end{bmatrix}$$

So that we can have

$$\begin{bmatrix} u_1' \\ u_2' \\ u_3' \\ u_4' \end{bmatrix} = \begin{bmatrix} u_2 \\ -GMu_1 / (u_1^2 + u_3^2)^{3/2} \\ u_4 \\ -GMu_3 / (u_1^2 + u_3^2)^{3/2} \end{bmatrix}$$

3.2 Problem 9.4

Consider the ODE $y' = -5y$ with initial condition $y(0) = 1$. We will solve this ODE numerically using a step size of $h = 0.5$.

(a) Are solutions to this ODE stable?

Solution. Solving the ODE, we have

$$\frac{dy}{dt} = -5y$$

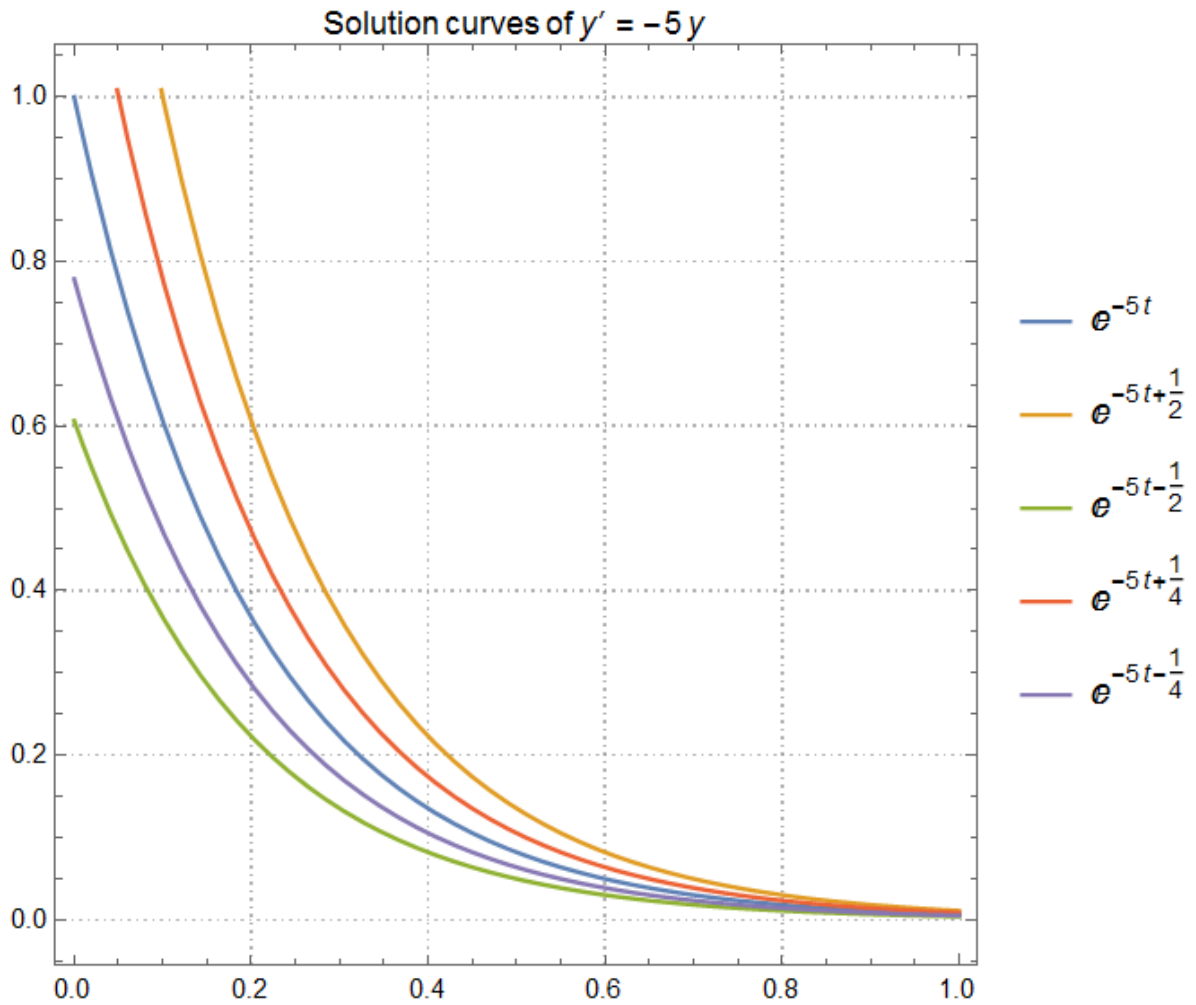
$$\frac{dy}{y} = -5 dt$$

$$\int \frac{dy}{y} = \int -5 dt$$

$$\ln y = -5t + C$$

$$y = e^{-5t+C}$$

Visualize the family of curves as solutions to this ODE.



Applying the initial condition, we have $y = e^{-5t}$, which is shown in blue. If the initial value is perturbed, then the perturbed solution remains close to the original solution. Moreover, notice that the solution curves $y \rightarrow 0$ as $t \rightarrow \infty$. Therefore, solutions to this ODE are asymptotically stable.

(b) Is Euler's method stable for this ODE using this step size?

Solution. In order for Euler's method to be stable in the ODE $y' = \lambda y$, the step size h must satisfy the inequality

$$\begin{aligned}
 |1 + h\lambda| &\leq 1 \\
 |1 + (0.5)(-5)| &\leq 1 \\
 |1 - 2.5| &\leq 1 \\
 1.5 &\not\leq 1
 \end{aligned}$$

It turns out that the Euler's method is not stable for the ODE given above.

(c) Compute the numerical value for the approximate solution at $t = 0.5$ given by Euler's method.

Solution. Euler's explicit method is given by

$$\begin{aligned}y_{k+1} &= y_k + h_k f(t_k, y_k) \\y_1 &= y_0 + (0.5)(-5y_0) \\y_1 &= 1 + (0.5)(-5 \cdot 1) \\y_1 &= -1.5\end{aligned}$$

(b) Is the backward Euler method stable for this ODE using this step size?

Solution. For the backward Euler method to be stable we must have

$$\begin{aligned}\left| \frac{1}{1 - h\lambda} \right| &\leq 1 \\ \left| \frac{1}{1 - (0.5)(-5)} \right| &\leq 1 \\ \left| \frac{1}{-1.5} \right| &\leq 1 \\ 0.\bar{6} &\leq 1\end{aligned}$$

Actually, this inequality holds for any $h > 0$ when $\text{Re}(\lambda) < 0$. Therefore, the backward Euler method is stable for this ODE.

(c) Compute the numerical value for the approximate solution at $t = 0.5$ given by the backward Euler method.

Solution. The backward Euler method is given by

$$\begin{aligned}y_{k+1} &= y_k + h_k f(t_{k+1}, y_{k+1}) \\y_1 &= y_0 + (0.5)(-5y_1) \\y_1 &= 1 - 2.5y_1 \\3.5y_1 &= 1 \\y_1 &= 0.285714 \dots\end{aligned}$$

3.3 Problem 9.8

Consider the IVP for the ODE $y' = -y^2$ with the initial condition $y(0) = 1$. We will use the backward Euler method to compute the approximate value of the solution y_1 at time $t_1=0.1$ (i.e. taking one step using the backward Euler method with step size $h = 0.1$ starting from $y_0 = 1$ at $t_0 = 0$). Since the backward Euler method is implicit, and the ODE is nonlinear, we will need to solve a nonlinear algebraic equation for y_1 .

(a) Write out that nonlinear algebraic equation for y_1 .

Solution. The backward Euler method is given by

$$\begin{aligned}y_{k+1} &= y_k + h_k f(t_{k+1}, y_{k+1}) \\y_1 &= 1 + (0.1)(-y_1^2)\end{aligned}$$

$$\boxed{(0.1)y_1^2 + y_1 - 1 = 0} \quad (22)$$

(b) **Write out Newton iteration for solving the nonlinear algebraic equation.**

Solution. The Newton's root-finding method is given by

$$\begin{aligned} y^* &= y - \frac{f(y)}{f'(y)} \\ f(y) &= (0.1)y^2 + y - 1 \\ f'(y) &= (0.2)y + 1 \\ \boxed{y^* &= y - \frac{(0.1)y^2 + y - 1}{(0.2)y + 1}} \end{aligned}$$

(c) **Obtain a starting guess for the Newton iteration by using one step of Euler's method for the ODE.**

Solution. Euler's explicit method is given by

$$\begin{aligned} y_{k+1} &= y_k + h_k f(t_k, y_k) \\ y_1 &= 1 + (0.1)(-y_0^2) \\ y_1 &= 1 + (0.1)(-1^2) \\ \boxed{y_1 &= 0.9} \end{aligned}$$

(d) **Finally, compute an approximate value for the solution y_1 by using one iteration of Newton's method for the nonlinear algebraic equation.**

Solution. Let's have one iteration:

$$\begin{aligned} y^* &= y - \frac{\frac{1}{10}y^2 + y - 1}{\frac{1}{5}y + 1} \\ y^* &= 0.9 - \frac{(0.1)(0.9)^2 + (0.9) - 1}{(0.2)(0.9) + 1} \\ y^* &= 0.9 - \frac{-0.019}{1.18} \\ y^* &= 0.9 + 0.0161017 \\ \boxed{y^* &= 0.9161017} \end{aligned}$$

Further Notes.

Solving the nonlinear equation (22) analytically, we have

$$\begin{aligned} (0.1)y^2 + y - 1 &= 0 \\ y &= \frac{-1 \pm \sqrt{1^2 - 4(0.1)(-1)}}{0.2} \\ y &= \frac{-1 \pm \sqrt{1.4}}{0.2} \\ \boxed{y_1 &= 0.9160797 \dots} \\ y_2 &= -10.9160797 \dots \end{aligned}$$

Notice that y^* is already close to the true value given in y_1 above.

3.4 Problem 9.9

For each property listed below, state which of the following three ODE methods has or have the given property.

$$(1) \ y_{k+1} = y_k + \frac{h}{2} (f(t_k, y_k) + f(t_{k+1}, y_k + hf(t_k, y_k)))$$

a, d, f

$$(2) \ y_{k+1} = y_k + \frac{h}{2} (3f(t_k, y_k) - f(t_{k-1}, y_{k-1}))$$

a

$$(3) \ y_{k+1} = y_k + \frac{h}{2} (f(t_k, y_k) + f(t_{k+1}, y_{k+1}))$$

a, d, e, c, g